

FATE: Monitoring DIC

2025 Winter Samples

2026-04-25

Contents

0.1	Import Data Functions	2
0.2	Import Sample Data	2
0.3	Assessing Standard Curves	3
0.4	CRMs Check: CRM Batch 221-0866	4
0.5	Assess Check Standards	5
0.6	Assess Blanks	6
0.7	Assess Duplicate Samples	7
0.8	Sample Flagging - Are samples Within the range of the curve?	8
0.9	Visualize Data by Dock & Flume & Date	9
0.10	Convert data from mg/L to uMoles/L	10
0.11	Export Processed Data	10

```
##Setup - Change things here & write any notes
```

```
#identify section  
cat("Setup Information")
```

```
## Setup Information
```

```
##### Run information - PLEASE CHANGE  
Date_Run = "02/02/26" #Date that instrument was run  
Run_by = "Stephanie J. Wilson" #Instrument user  
Script_run_by = "Stephanie J. Wilson" #Code user  
run_notes = "NA" #any notes from the run  
samples <- c("FATE") #whatever identifies your samples within the same names  
samples_pattern <- paste(samples, collapse = "|")  
  #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code  
chks_name = "Chk_Std_" #what did you name your check standards?  
crm_name = "CRM|crm" #what did you name your CRMS?  
  
##### File Names - PLEASE CHANGE  
#file path and name for raw summary data file  
raw_file_name = "Raw Data/FATE_DIC_202601.txt"  
  
#file path and name for raw all peaks file  
raw_allpeaks_name = "Raw Data/FATE_DIC_202601_allpeaks.txt"  
  
#file path and name of processed data file  
processed_file_name = "Processed Data/FATE_Processed_DIC_202601.csv"  
  
##### Log Files - PLEASE CHECK  
#downloaded metadata csv - downloaded from Google drive as csv for this year  
  #Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv"  
  
#qaqc log file path for this year  
Log_path = "Raw Data/FATE_DIC_QAQClog_2025.csv"
```

```
##Set Up Code
```

0.1 Import Data Functions

0.2 Import Sample Data

```
## Import Sample Data
```

```
## New names:  
## * ' ' -> '...14'
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,  
## e.g.:  
##   dat <- vroom(...)  
##   problems(dat)
```

```
## # A tibble: 4 x 3
##   sample_name      ic_raw run_datetime
##   <chr>          <dbl> <chr>
## 1 FATE_Flume_20251204_A    20.8 2/2/2026 5:46:47 PM
## 2 FATE_Flume_20251204_B    22.4 2/2/2026 5:57:56 PM
## 3 FATE_Dock_20251204_A     19.5 2/2/2026 6:08:57 PM
## 4 FATE_Dock_20251204_B     19.4 2/2/2026 6:20:15 PM
```

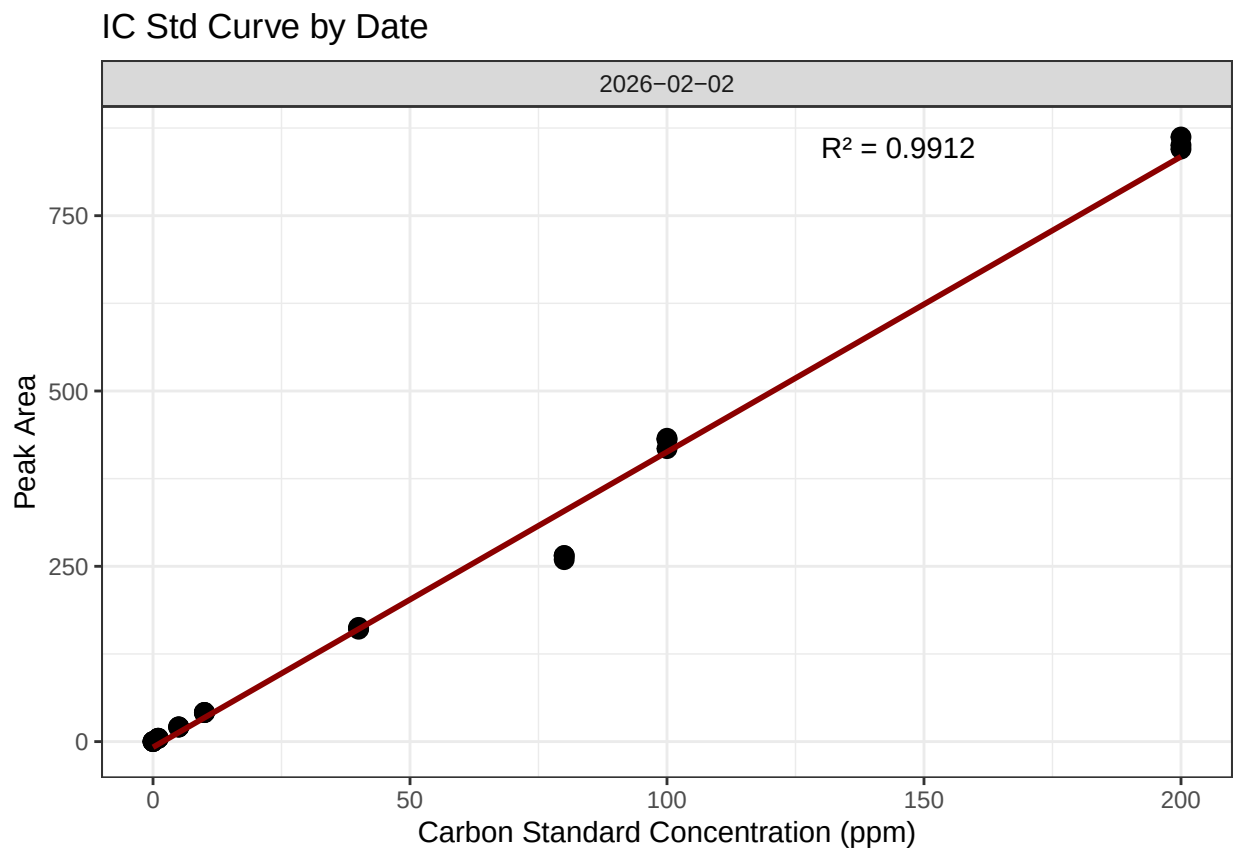
0.3 Assessing Standard Curves

```
## Assess the Standard Curves
```

```
## New names:
## * ' ' -> '...18'
```

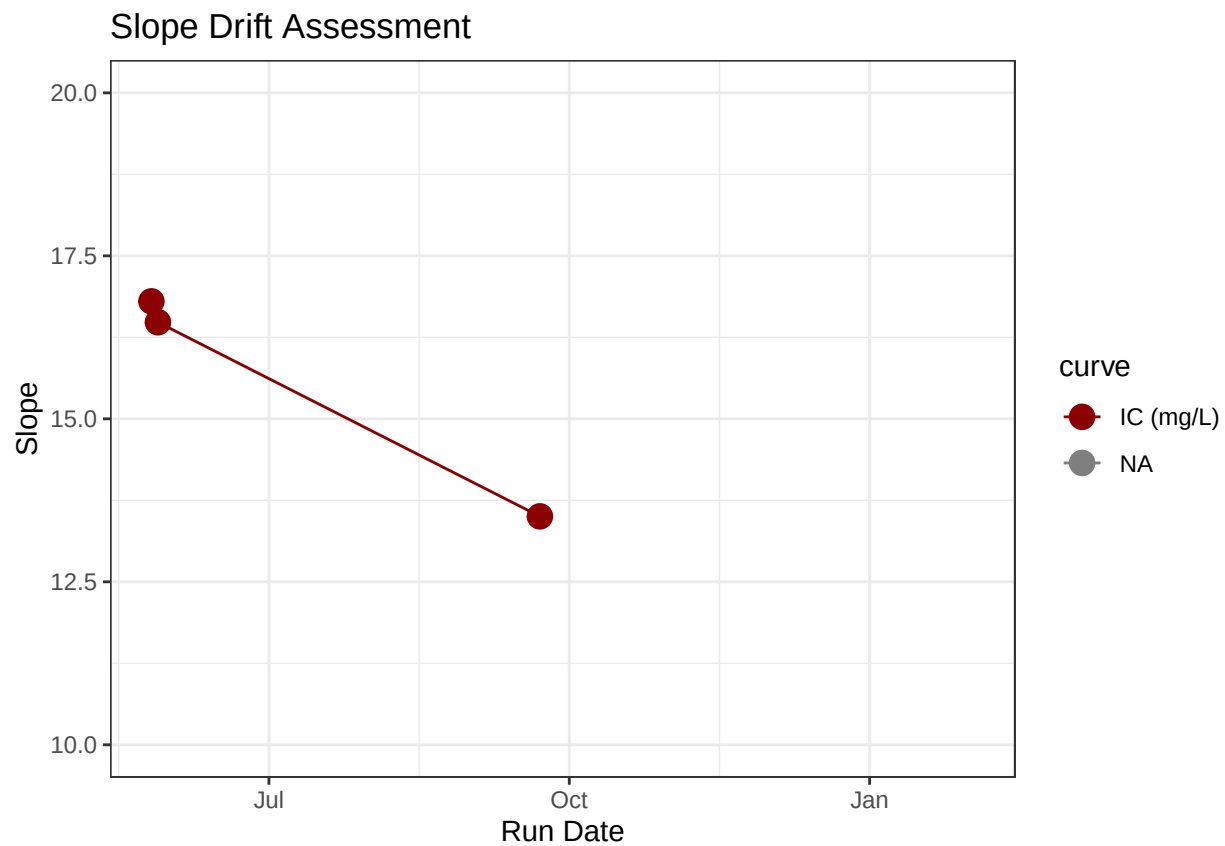
```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## ('geom_line()').
```



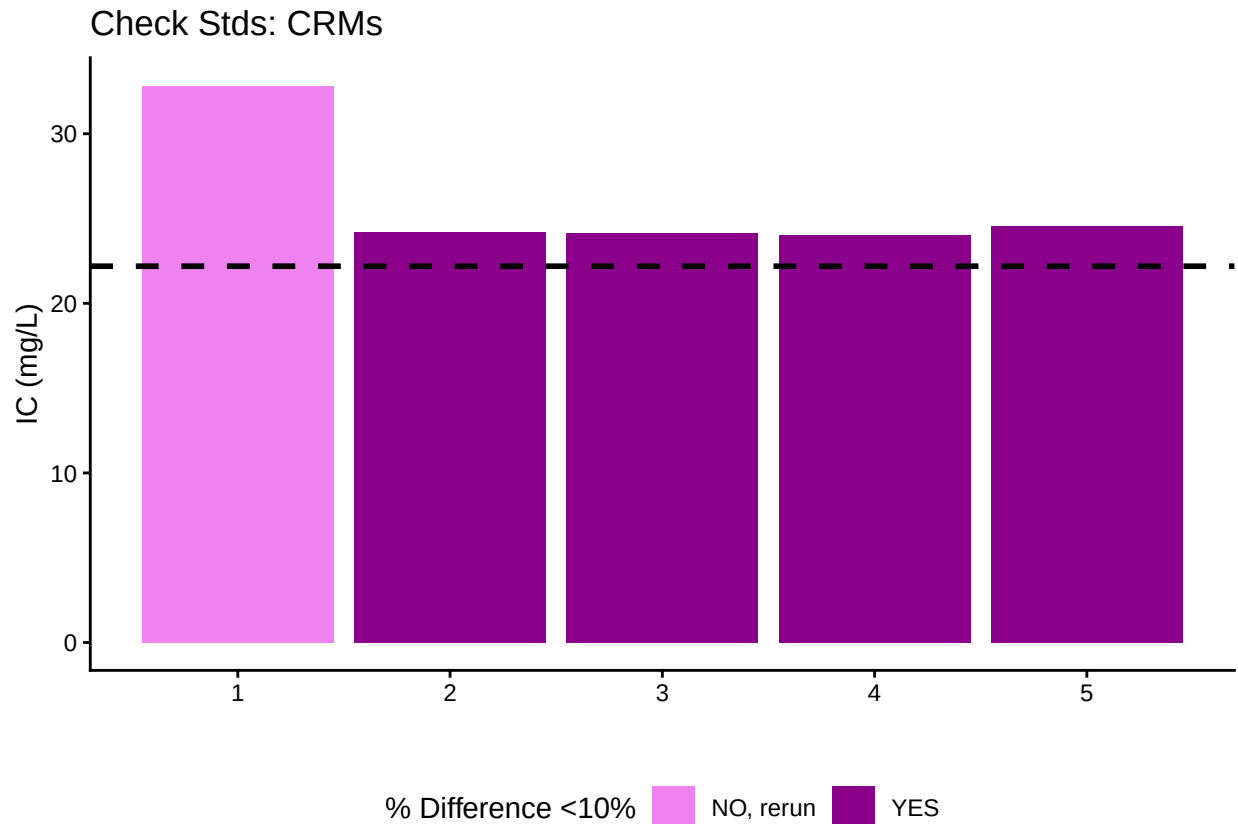
```
## [1] "IC Curve r2 GOOD"
```

0.4 CRMs Check: CRM Batch 221-0866

```
## Assess the CRMs
```

```
## New names:
## * ' ' -> '...14'
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```



```
## [1] "IC crm has a % Difference >25% of expected - REASSESS"
```

```
## Run mean = 25.964
```

```
## Expected = 22.19
```

0.5 Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
```

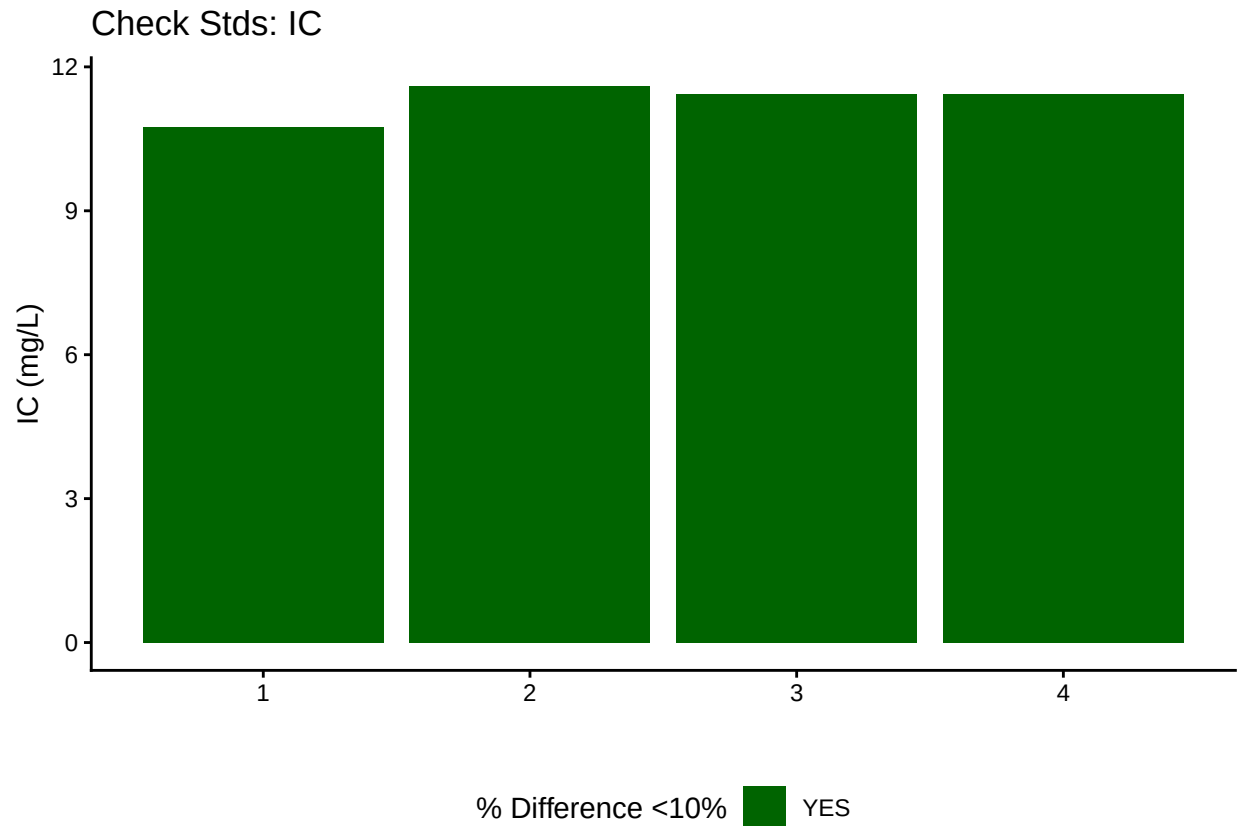
```
## * ' ' -> '...14'
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
```

```
## e.g.:
```

```
## dat <- vroom(...)
```

```
## problems(dat)
```



```
## [1] ">60% of IC Check Standards are within range of expected concentration"
```

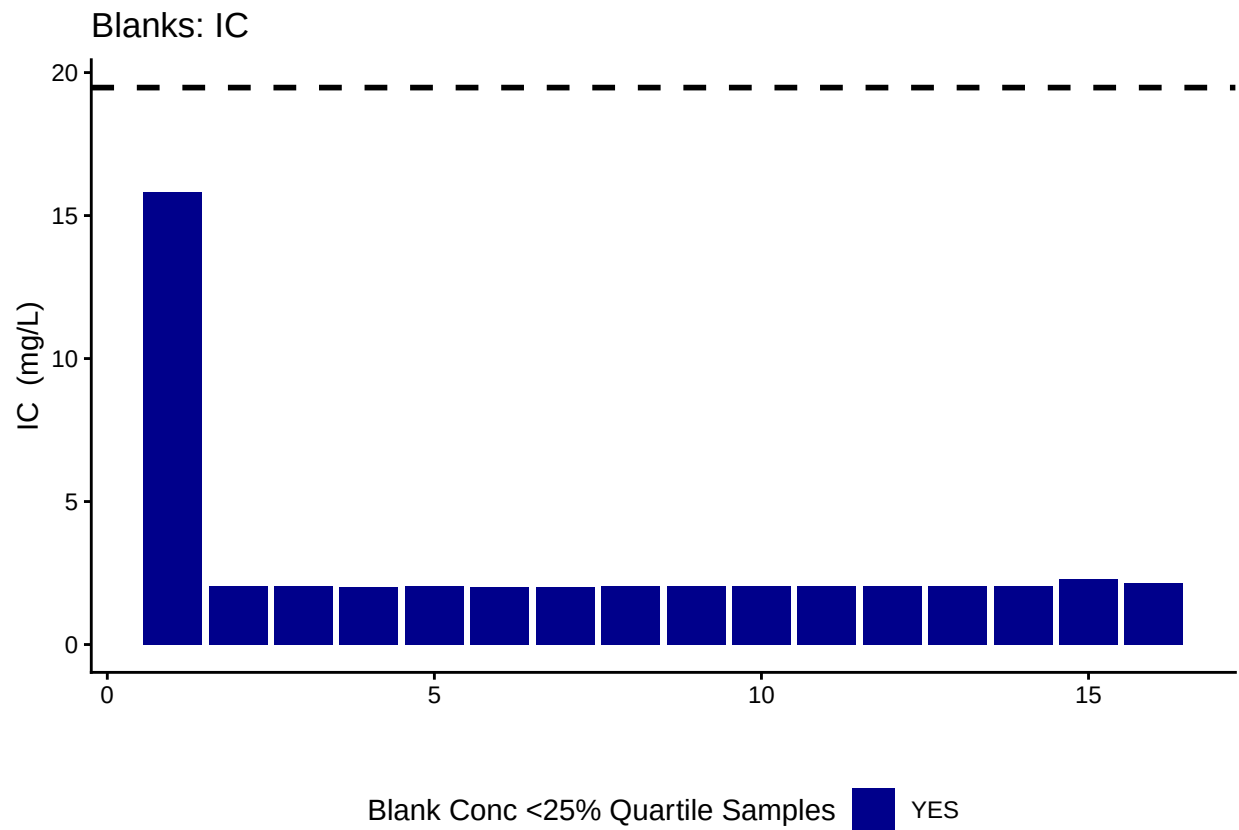
0.6 Assess Blanks

```
## Assess Blanks
```

```
## New names:
## * ' ' -> '...14'
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

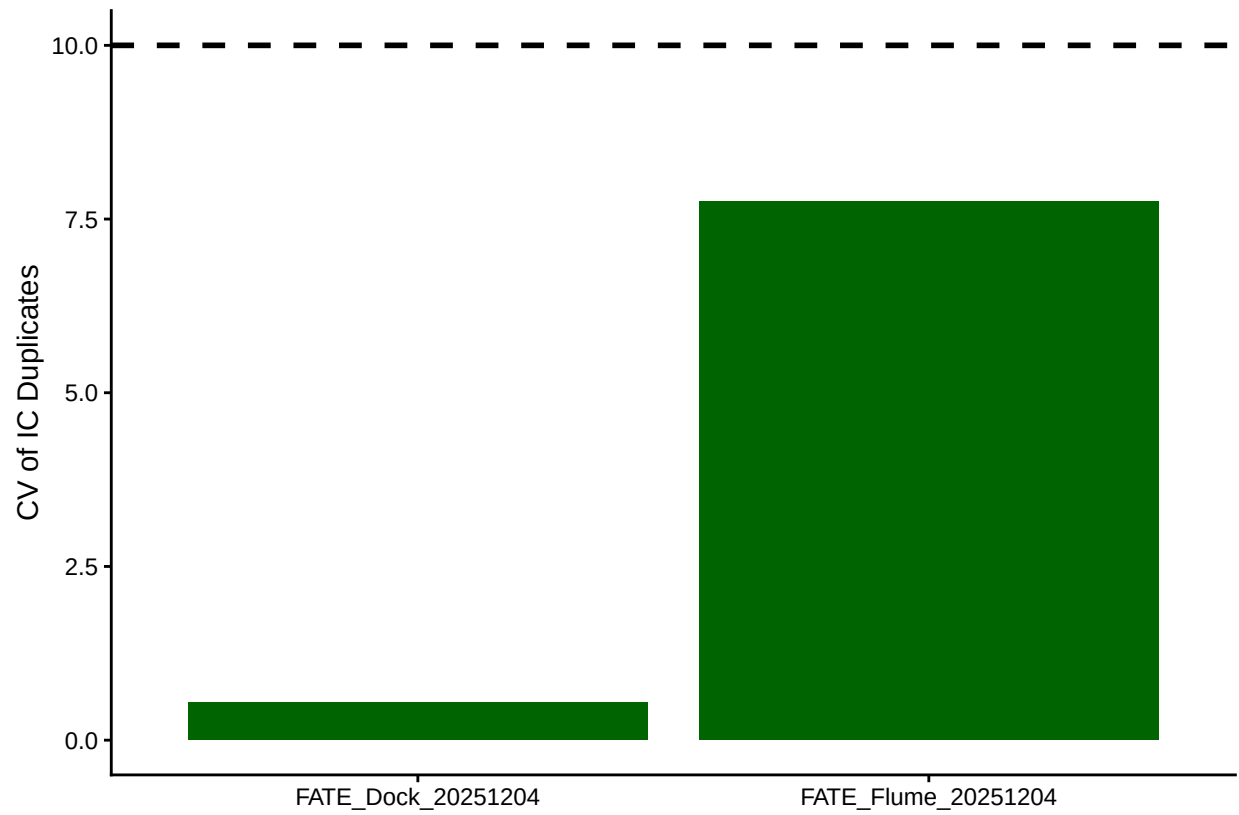


```
## carbon blanks:
```

```
## [1] 2.9185
```

0.7 Assess Duplicate Samples

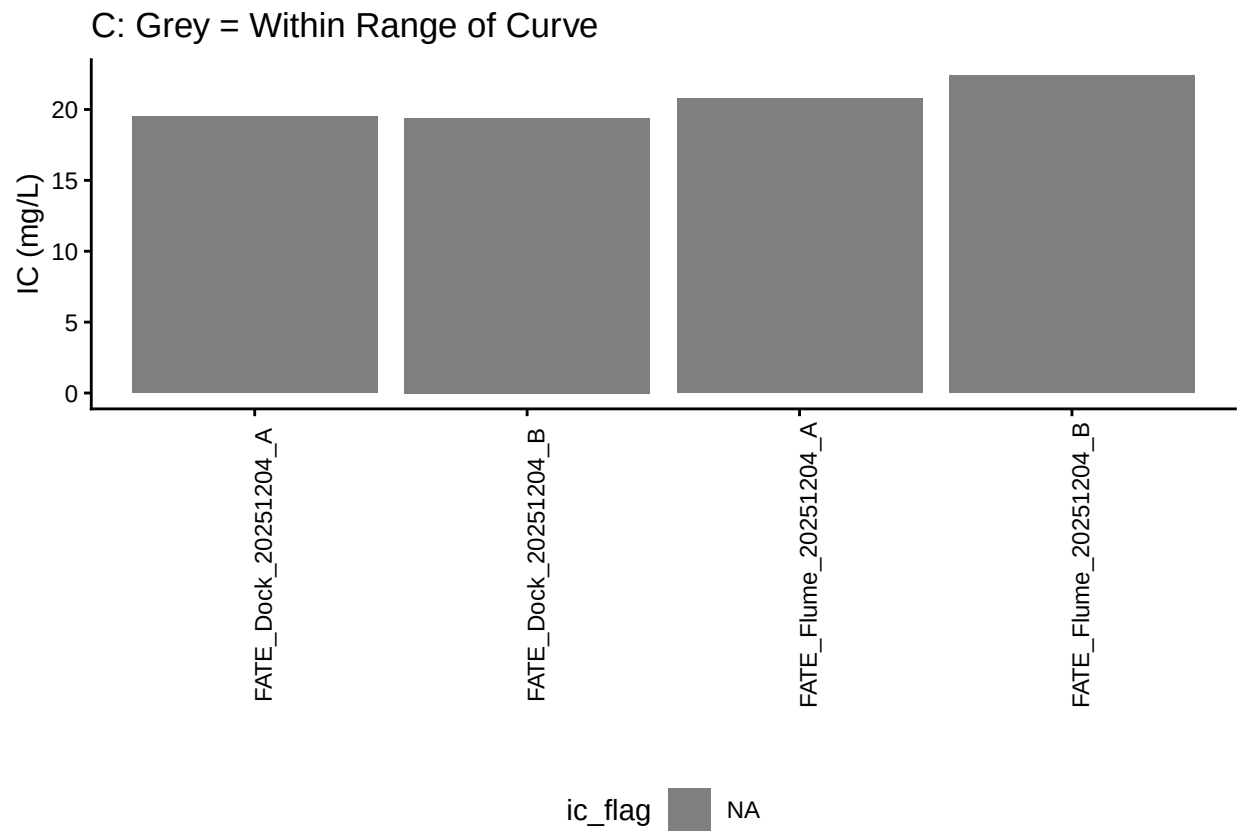
```
## Assess Duplicates
```



```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

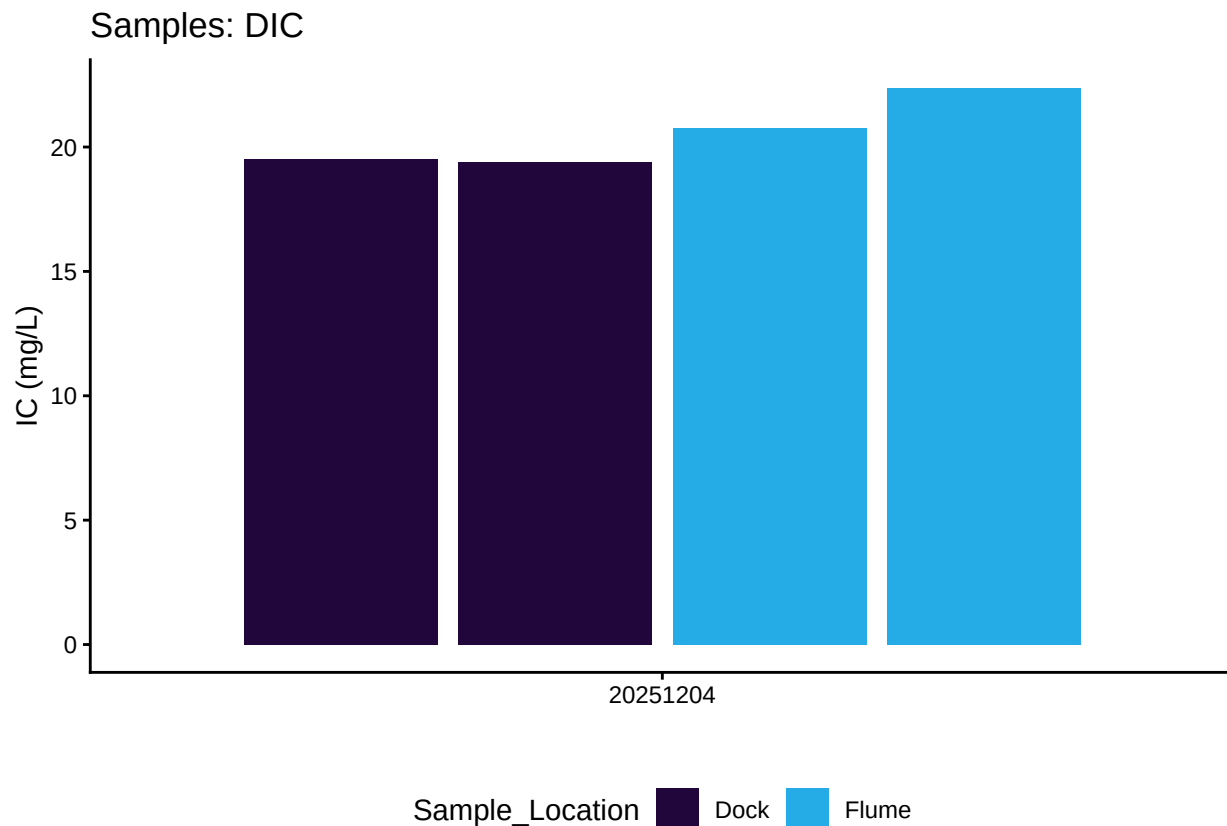
0.8 Sample Flagging - Are samples Within the range of the curve?

```
## Sample Flagging
```

0.9 Visualize Data by Dock & Flume & Date

Visualize Data



0.10 Convert data from mg/L to uMoles/L

0.11 Export Processed Data

Export Processed Data

```
## # A tibble: 4 x 9
##   Project Sample_Location Replicate Date   ic_mgL ic_uM ic_flag Analysis_runtime
##   <chr>    <chr>           <chr>   <chr>   <dbl> <dbl> <chr>   <chr>
## 1 FATE    Flume             A      20251~  20.8 1729. ""      2/2/2026 5:46:4~
## 2 FATE    Flume             B      20251~  22.4 1864. ""      2/2/2026 5:57:5~
## 3 FATE    Dock              A      20251~  19.5 1624. ""      2/2/2026 6:08:5~
## 4 FATE    Dock              B      20251~  19.4 1615. ""      2/2/2026 6:20:1~
## # i 1 more variable: Run_notes <chr>
```

#end